UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

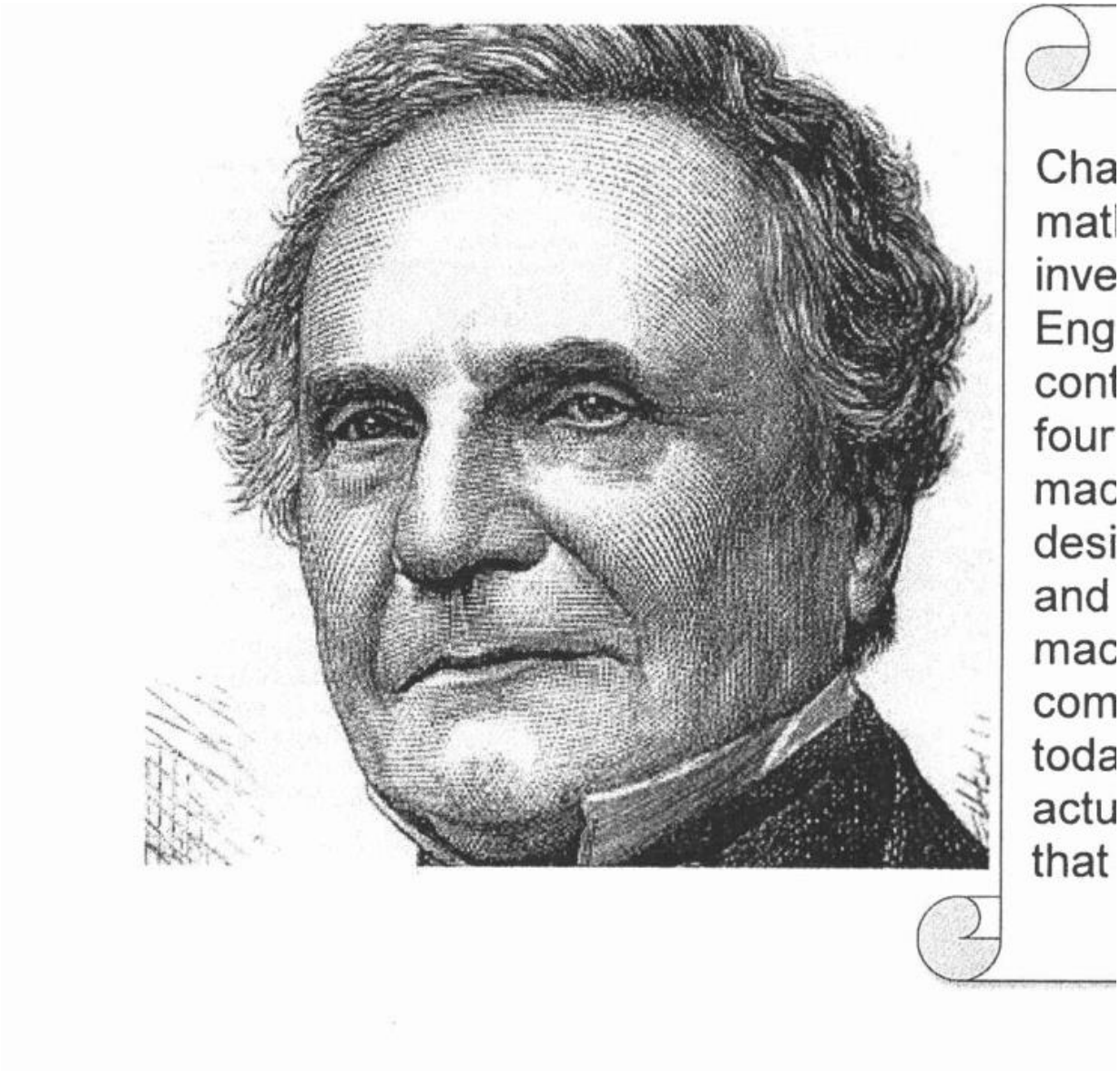
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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

Question 1 is based on the following image and statements in Figure 1.



Charles Babbage was an English mathematician, a philosopher and also an inventor. He was born on 1791 in London England. He was known for his contributions to invent machine X to solve four basic arithmetic functions. The machine X has improved its calculations design from Stepped Reckoner machine and early mechanical calculator. The machine X has introduced a number of computing concepts which are still in use today. Although this machine was not actually completed, it is the first machine that deserved to be called as a computer.

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a) What is the machine X name?

b) In which generation does the machine X introduced?

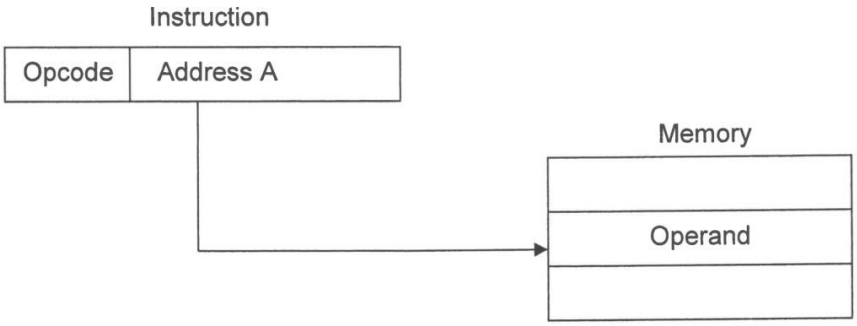
c) What is the function of machine X?

d) Name the machine that has been invented prior to machine X?

e) What is the title given to Charles Babbage due to his success to invent machine X?

QUESTION A2 **(5 marks)**

Given the following Figure 2.



a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

QUESTION A3 **(5 marks)**

a) List THREE (3) characteristics of Symmetric Multiprocessor (SMP).

b) Describe ONE (1) advantage of Symmetric Multiprocessor (SMP) interms of availability.

QUESTION A4 **(5 marks)**

Convert 378510 to binary. Show your complete workflow.

QUESTION A5 **(10 marks)**

Answer both parts for the same expression. Assume an 8-bit byte.

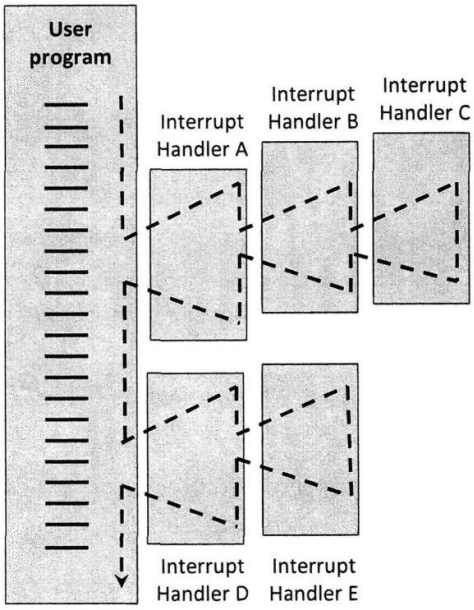
Expression: -60 - (-12)

a) Calculate -60 - (-12) using sign and magnitude.

b) Calculate -60 - (-12) using 1's and 2's complement.

QUESTION A6 **(10 marks)**

Question 8 is based on Figure 2.



a) Describe maskable interrupt.

b) What is the sequence of the program execution if the defined priority is implemented in

c) What is the sequence of the program execution if the disable interrupt is implemented in

d) Which of the interruption method between defined priority or disable interrupt can finish Interrupt C earlier? Give your justification.

QUESTION A7 **(10 marks)**

Describe the specific function of each registers in the CPU.

QUESTION A8 **(10 marks)**

a) Discuss the function of each Central Processing Unit (CPU) component'

b) Elaborate on what will happen in the machine cycle if the bus system is faulty

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QUESTION A9 **(10 marks)**

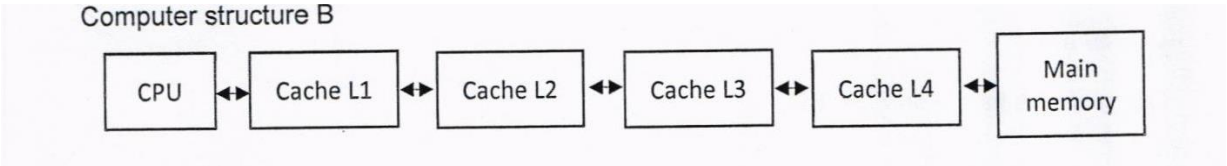
a) Differentiate between interrupt-driven 1/0 and programmed 1/0.

b) Discuss ONE (1) benefit of programmed 1/0.

c) Discuss TWO (2) disadvantages of multiple interrupts.

QUESTION A10 **(10 marks)**

Question I is based on Figure 3. Computer structure A Main cache L2 Cache L3 CPU cache L1 memory



a) Explain which computer structure can store the data faster compared to the other. Provide relevant justification to support your answer.

b) Identify the fastest cache level and give an appropriate reason.

c) Vvhat is the main benefit for the cache to use SRAM technology?

d) Discuss TWO (2) methods to update cache and main memory contents to ensure data consistency.

PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 2 and Table 1.

$$y = A - \frac{2C}{B + A^2} * D$$

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three- address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

One-address	Two-address	Three-address
LOADp	MOV (p,q)	ADD (p.q.r)
STORp	ADD (p,q)	SUB (p,q,r)
ADDp	SUB (p,q)	MUL (p.q.r)
SUBp	MUL (p,q)	DIV(p,q,r)
MULp	DIV(p,q)	
DIVp		

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